# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2000-I-Q05] | Part A (Q1-Q5)

## Answer: 026

1. Let box sizes be $a, b$ and black counts $p, q$. Then $a + b = 25$ and $50pq = 27ab$, so $50 \mid ab$. Up to exchange, only $(a, b) = (5, 20), (10, 15)$ fit.

2. For sizes $5, 20$, $pq = 54$ with $p < 5, q < 20$ forces black counts $3, 18$, leaving whites $2, 2$. For sizes $10, 15$, $pq = 81$ forces $9, 9$, leaving $1, 6$.

3. Both cases give white probability $1/25$, so the answer is $26$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2000-II-Q03] | Part A (Q1-Q5)

## Answer: 758

1. The remaining multiplicities are nine groups of four and one group of two, so there are $9\binom{4}{2}+\binom{2}{2}=55$ matching pairs.

2. There are $\binom{38}{2}=703$ possible pairs. Since $55/703$ is reduced, the answer is $758$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2002-I-Q01] | Part A (Q1-Q5)

## Answer: 059

1. In a three-symbol block, being palindromic means the last symbol equals the first. The two block probabilities are $1/26, 1/10$, independently.

2. The union probability is $1/26 + 1/10 - 1/260 = 7/52$, so the sum is $59$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2005-II-Q02] | Part A (Q1-Q5)

## Answer: 079

1. For the first bag, after any initial roll, the next differs in type with probability $6/8$ and the third supplies the missing type with probability $3/7$. Thus the chance is $9/28$.

2. Two of each type remain. The second bag succeeds with chance $(4/5)(2/4) = 2/5$, and the last bag is then forced. Probability $9/70$, giving $79$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2010-I-Q01] | Part A (Q1-Q5)

## Answer: 107

1. $2010^2 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 67^2$ has $3^4 = 81$ positive divisors. The square divisors choose each exponent from $0, 2$, so there are16 squares and65 nonsquares.

2. The probability is $16 \cdot 65 / \binom{81}{2} = 26/81$, giving $107$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2010-II-Q04] | Part A (Q1-Q5)

## Answer: 052

1. There are $12 \cdot 11$ ordered distinct gate pairs. For a separation of $d$ positions, there are $2(12-d)$ pairs.

2. Add $d = 1, 2, 3, 4$: favorable pairs total $2(11+10+9+8) = 76$. The probability is $76/132 = 19/33$.

3. Thus $m + n = 52$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2013-I-Q04] | Part A (Q1-Q5)

## Answer: 429

1. Each four-cell orbit must be monochromatic. Since the total red count is eight, the central cell is blue and exactly two of the three orbits are red.

2. There are three favorable choices and $\binom{13}{8} = 1287$ total colorings. Thus the probability is $3/1287 = 1/429$, giving $429$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2015-I-Q02] | Part A (Q1-Q5)

## Answer: 139

1. There are $\binom{9}{3} = 84$ possible subsets. Choosing the nationality of the repeated pair gives favorable count $\binom{2}{2} \cdot 7 + \binom{3}{2} \cdot 6 + \binom{4}{2} \cdot 5 = 55$.

2. The cases are disjoint because a three-person subset cannot have two different repeated nationalities. Thus the probability is $55/84$, and $m + n = 139$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2015-II-Q05] | Part A (Q1-Q5)

## Answer: 090

1. The grid has $n(n-1)$ horizontal and the same number of vertical neighboring pairs. Among $\binom{n^2}{2}$ unordered pairs, the probability is $2n(n-1)/\binom{n^2}{2} = 4/[n(n+1)]$.

2. The strict inequality becomes $n(n+1) > 8060$. Since $89 \cdot 90 = 8010 < 8060 < 90 \cdot 91 = 8190$, the least possible $n$ is $90$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2018-I-Q03] | Part A (Q1-Q5)

## Answer: 157

1. Treat individual cards as distinct, giving $10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 = 30240$ equally likely layouts. For a fixed color word with $r$ reds, the number of fillings is $(5!/(5-r)!)(5!/r!)$.

2. The monochromatic words give $120$ each. For $r = 1, 2, 3, 4$, two block orders qualify, with respective fillings $600, 1200, 1200, 600$.

3. Favorable layouts total $240 + 2(3600) = 7440$, so probability $31/126$ and answer $157$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2019-I-Q02] | Part A (Q1-Q5)

## Answer: 029

1. The $20 \cdot 19 = 380$ ordered pairs are equally likely. Favorable choices for $J = 1, \ldots, 18$ number $19 - J$.

2. Sum $18 + \cdots + 1 = 171$, so probability $171/380 = 9/20$, giving $29$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2019-I-Q05] | Part A (Q1-Q5)

## Answer: 252

1. Success requires reaching $(1,1)$ and then taking a diagonal. If the preceding path has $d$ diagonals, it has $3-d$ steps of each other type and length $6-d$. Their multinomial counts for $d=0,1,2,3$ are $20,30,12,1$.

2. Weight paths by their lengths and the final diagonal: probability $\frac{1}{3}(20/3^6+30/3^5+12/3^4+1/3^3)=245/3^7$. Thus $m+n=252$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2023-I-Q01] | Part A (Q1-Q5)

## Answer: 191

1. Treat the seven diameters as pairs of positions. The five male positions are uniformly distributed among $\binom{14}{5} = 2002$ subsets.

2. Choose five distinct diameter pairs, then one end of each: $\binom{7}{5}2^5 = 672$ choices.

3. Thus the probability is $672/2002 = 48/143$, and $m + n = 191$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2026-II-Q05] | Part A (Q1-Q5)

## Answer: 190

1. Equal hypergeometric counts give $\binom{r}{4}\binom{b}{3}=\binom{r}{5}\binom{b}{2}$. Cancel to obtain $(b-2)/3=(r-4)/5$, or $5b=3r-2$.

2. Thus $r\equiv 4 \pmod 5$. The bounds make the first five $r$ values $14,19,24,29,34$, with $b=8,11,14,17,20$.

3. Totals are $22,30,38,46,54$, whose sum is $190$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q06] | Part B (Q6-Q10)

## Answer: 079

1. Each multiset of four values from six labels has exactly one nondecreasing ordering. The favorable count is $\binom{9}{4} = 126$ by weak compositions of four into six multiplicities.

2. Of $6^4 = 1296$ equally likely outcomes, the probability is $126/1296 = 7/72$, giving $79$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q10] | Part B (Q6-Q10)

## Answer: 200

1. The midpoint automatically stays in the geometric box; it is a lattice point exactly when the coordinate parities agree. Allowing equal ordered points, the same-parity coordinate pair counts are $2^2+1^2=5$, $2^2+2^2=8$, $3^2+2^2=13$.

2. Of their $520$ product pairs, $60$ are equal-point pairs. Therefore the desired probability is $460/(60\cdot 59)=23/177$, giving $200$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-II-Q09] | Part B (Q6-Q10)

## Answer: 929

1. For the four block events, a single block forces $4$ red cells. Of six block pairs, four force $6$ cells and two force 7. Each triple forces $8$ cells; all four force $9$.

2. In units of $1/512$, the count containing a red block is $4 \cdot 32 - (4 \cdot 8 + 2 \cdot 4) + 4 \cdot 2 - 1 = 95$ .

3. The desired probability is $(512 - 95)/512 = 417/512$, giving $929$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2004-II-Q10] | Part B (Q6-Q10)

## Answer: 913

1. Each number is $2^a + 2^b$ for $0 \le a < b \le 39$, giving $\binom{40}{2} = 780$ choices.

2. Divisibility by nine is equivalent to $2^{b-a} \equiv -1 \pmod 9$, hence $b - a \equiv 3 \pmod 6$. Allowed differences $3, 9, \ldots, 39$ contribute $37 + 31 + 25 + 19 + 13 + 7 + 1 = 133$ pairs.

3. The probability $133/780$ is reduced, giving $913$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2005-I-Q09] | Part B (Q6-Q10)

## Answer: 074

1. The edge joining the two unpainted faces is uniformly one of twelve edges. At a large-cube corner, edge position, face center, the fractions placing both bad faces inward are $3/12 = 1/4$, $5/12$, and $8/12 = 2/3$, respectively.

2. With eight corner, twelve edge and six face-center cubes, multiply to get $(1/4)^8(5/12)^{12}(2/3)^6 = 5^{12}/(2^{34}3^{18})$. The central cube imposes no condition.

3. The requested sum is $12 + 34 + 18 + 5 + 2 + 3 = 74$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2007-II-Q10] | Part B (Q6-Q10)

## Answer: 710

1. Given $|A| = j$, allowed $B$ number $2^j + 2^{6-j} - 1$, subtracting the empty-set overlap.

2. Summing over $A$ gives favorable pairs $\sum_j \binom{6}{j}(2^j + 2^{6-j} - 1) = 2 \cdot 3^6 - 2^6$. Divide by $2^{12}$ to get $697/2^{11}$, so the answer is $710$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2011-I-Q10] | Part B (Q6-Q10)

## Answer: 503

1. An inscribed triangle is obtuse exactly when all three vertices lie in an open semicircle. Its unique first clockwise vertex determines the other two within that half-circle, so this counts each obtuse triangle once.

2. For $n = 2k$, there are $n\binom{k-1}{2}$ obtuse triples. Dividing by $\binom{2k}{3}$ gives $3(k-2)/[2(2k-1)] = 93/125$, whose root is $k = 188$, or $n = 376$.

3. For $n = 2k+1$, the probability is $3(k-1)/[2(2k-1)] = 93/125$, giving $k = 63$, or $n = 127$.

4. Both satisfy the count formulas; the required sum is $503$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2013-I-Q06] | Part B (Q6-Q10)

## Answer: 047

1. The three math-book positions form a uniformly random three-subset of twelve slots. A box of capacity $k$ contains them in $\binom{k}{3}$ possible slot subsets.

2. The favorable counts add because boxes are disjoint: $\binom{3}{3}+\binom{4}{3}+\binom{5}{3}=15$. Thus probability $15/\binom{12}{3}=3/44$, giving $47$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2020-I-Q09] | Part B (Q6-Q10)

## Answer: 077

1. Every divisor is uniquely $2^u5^v$, with $u \in \{0, \ldots, 18\}$ and $v \in \{0, \ldots, 9\}$. Both exponent coordinates are independent uniform choices.

2. For $k$ possible exponents, a nondecreasing triple has $\binom{k+2}{3}$ possibilities. Consequently the probability is $\binom{21}{3}\binom{12}{3}/190^3$.

3. This equals $(70/361)(11/50) = 77/1805$. As $77$ and $1805$ are coprime, the numerator is $77$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2022-I-Q09] | Part B (Q6-Q10)

## Answer: 247

1. For positions $i < j$, the gap count $j - i - 1$ is even exactly when $i$ and $j$ have opposite parity. Thus every color must occur once in the six odd slots and once in the six even slots.

2. There are $(6!)^2$ favorable color rows and $12!/2^6$ total color rows. Their ratio is $2^6(6!)^2/12! = 16/231$.

3. The numerator and denominator are coprime, giving $m + n = 247$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2025-I-Q07] | Part B (Q6-Q10)

## Answer: 821

1. The total pairings are $11!! = 10395$. The last word has the largest smaller letter among all pairs.

2. If $G$ is the first letter, it has five later partners. Each remaining later letter must pair with a distinct member of $A, \ldots, F$, giving $5 \cdot (6 \cdot 5 \cdot 4 \cdot 3) = 1800$ pairings.

3. If $G$ is the second letter, its partner must be $F$: otherwise too many later letters would require earlier partners. The five remaining late letters biject with $A, \ldots, E$, giving $120$ pairings. Probability $(1800 + 120)/10395 = 128/693$, so the answer is $821$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2001-I-Q15] | Part C (Q11-Q15)

## Answer: 085

1. The dual graph is a cube. Encode its vertices by binary triples. Two successive labels are allowed exactly when their vertices differ in two or three bits.

2. Fix label $1$ at vertex $000$. Let $H(S, v)$ count paths starting there, using exactly subset $S$, ending at $v$, with every step allowed. Start $H(\{000\}, 000) = 1$, and append unused $w$ when the bit-distance from $v$ to $w$ is at least two.

3. The sums of these path counts at lengths $1$ through $8$ are $1, 4, 12, 30, 66, 126, 156, 102$. Of the full paths, $60$ have last vertex also allowed adjacent to $000$, closing the label cycle. The conditional favorable count is $60$ out of $7! = 5040$.

4. The probability is $1/84$, and the requested sum is $85$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2011-II-Q12] | Part C (Q11-Q15)

## Answer: 097

1. Country patterns are equally likely, numbering $9!/(3!)^3 = 1680$. Failure occurs exactly when a country occupies three consecutive seats, since its middle delegate then has two compatriot neighbors.

2. For one specified country, there are $9\binom{6}{3} = 180$ such patterns. For two specified countries, place the first block in nine ways and the second in one of four triples along the remaining six-seat arc: $36$ patterns. For all three, there are three cyclic partitions into triples and $3!$ country assignments, or $18$.

3. Inclusion-exclusion gives $3(180) - 3(36) + 18 = 450$ failures, hence probability $(1680 - 450)/1680 = 41/56$.

4. The answer is $97$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2014-I-Q12] | Part C (Q11-Q15)

## Answer: 453

1. If $f$ has image size $k$, exactly $(4-k)^4$ functions $g$ avoid it. Count $f$ by image size: $4$ for $k=1$; $\binom{4}{2}(2^4-2)=84$ for $k=2$; $\binom{4}{3}(3^4-3\cdot 2^4+3)=144$ for $k=3$; and $24$ for $k=4$.

2. Favorable ordered pairs number $4\cdot 81+84\cdot 16+144=1812$, out of $4^8=65536$.

3. The reduced probability is $453/16384$, giving $m=453$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2014-II-Q13] | Part C (Q11-Q15)

## Answer: 028

1. Pairings are permutations of owners: left shoe $j$ goes to right shoe $\pi(j)$. A collection using exactly its owners full pairs is an invariant subset of $\pi$, hence a union of cycles.

2. Thus there may be no cycle shorter than five. The only partitions of ten into allowable cycle lengths are ten or five plus five.

3. The favorable counts are $9!$ and $\binom{10}{5}(4!)^2/2 = 9!/5$. Dividing their sum by $10!$ gives $3/25$.

4. The requested sum is $28$.

Source / 来源

# Section 5.1 Finite probability & combinatorial sampling

有限概率与组合抽样

[AIME-2018-I-Q12] | Part C (Q11-Q15)

## Answer: 683

1. Six elements have each residue modulo three. Multiples of three may be chosen freely and cancel from the probability. Let $a, b$ be counts chosen from residues one and two; the condition is $a \equiv b \pmod 3$.

2. The cases $a - b = 0, \pm 3, \pm 6$ contribute $\binom{12}{6} = 924$, $2\binom{12}{9} = 440$, and two choices respectively, by Vandermonde convolution.

3. The probability is $1366/2^{12} = 683/2048$, so $m = 683$.

Source / 来源